

Using Satellite Data to Study Pollution

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Abstract

The goal of this paper is to investigate how remote sensing techniques can be utilized to assess environmental policies related to pollution. To study this topic, I begin by reviewing the underlying principles of using satellite instruments to measure pollution, which commonly involve the construction of indices for particulate matter in the atmosphere. Next, I discuss a few examples of academic papers that have benefited from the accessibility of publicly available satellite data, two of which aim to study the impact of airborne pollution on infant mortality. Lastly, I attempt to evaluate the effects of the disbandment of a high occupancy vehicle (HOV) policy in Jakarta, Indonesia using satellite data, with the goal being to compare the results to those produced using measures of pollution from ground monitoring sources. However, issues with missing values in my data and small sample size severely limit my analysis, highlighting potential pitfalls of using satellite data to study environmental policies.

1 Introduction

Air pollution is characterized as the presence of contaminants in the atmosphere, such as dust, smoke, or gas, in quantities that adversely affect human health (World Health Organization (2025)). Inhalation of such pollutants can lead to severe health outcomes such as stroke, heart disease, and lung cancer. It has also been associated with an increased risk of diabetes, neurological disease, cognitive impairment, and pregnancy complications such as low birth weight. Some important considerations in studying the harms of air pollution worldwide are its distributional effects. In 2016, the World Health Organization estimated that 7 million deaths worldwide could be attributed to air pollution, most of which occurred in developing countries (UNEP (2019)). This disparity is most likely due to a lack of stringent emission standards in such countries or a lack of enforcement of such policies, assuming that they do exist. Thus, it is of interest to social scientists to investigate where pollution is causing harm and to gain an understanding of which policy interventions have had success in mitigating its prevalence.

The advent of publicly available satellite imagery, coupled with remote sensing techniques, has greatly improved the ability of social scientists to study the

health impacts of pollution or the effectiveness of pollution-based policies at relatively low costs. In the past, researchers interested in evaluating pollution-based policies may have been constrained by the transparency of local governments, whose reporting of ground-based pollution measures can be subject to manipulation (Donaldson and Storeygard (2016)). Perhaps more importantly, the scarcity of pollution data for developing countries severely limits researchers' ability to study the health impacts of poor air quality in these areas. Ideally, researchers interested in air quality would collect pollution data from ground monitoring sources; however, even when developing countries have access to these data, their coverage tends to be limited to larger metropolitan areas and therefore may not be representative of other regions (Foster, Gutierrez, and Kumar (2009)). With remote sensing techniques, it is possible to bypass the constraints listed above and generate estimates of pollution levels throughout the entire world. This allows researchers to easily construct datasets containing remotely-sensed measures of pollution at many points in time, which lend themselves well to conventional econometric research designs.

The primary goal of this paper is to explore how access to satellite data has facilitated social science research on the impacts of environmental policies or phenomena related to airborne pollution. A secondary goal is to investigate whether satellite pollution measures can be used to replicate findings produced using ground monitoring sources. The outline of the paper is the following: Section II provides an overview of the principles of using remote sensing for measuring pollution and lists a few commonly used data sources, section III is a literature review, section IV focuses on my attempt to evaluate a real policy's impact on pollution, and section V is a conclusion.

2 Overview of Remotely-Sensed Pollution Data

2.1 Measuring Pollution with Satellite Instruments

A common method of estimating pollution levels with satellite instruments is by constructing measures of aerosol optical depth (AOD), which essentially estimate the prevalence of particulate matter in the atmosphere. This is achieved via sensors that record the electromagnetic radiation reflected from these particles, which are called aerosols (Foster, Gutierrez, and Kumar (2009)). While my literature review primarily focuses on the use of AOD in studies on pollution or the effects of pollution-based policies, it is certainly not the only remotely-sensed measure worth considering when studying these topics, as many satellite instruments can be used to estimate measures of other major pollutants such as carbon monoxide or nitrogen dioxide.

2.2 Common Sources of Pollution Data

One common source of AOD data is the Moderate Resolution Imaging Spectroradiometer, or MODIS, an instrument that flies onboard NASA's Aqua and

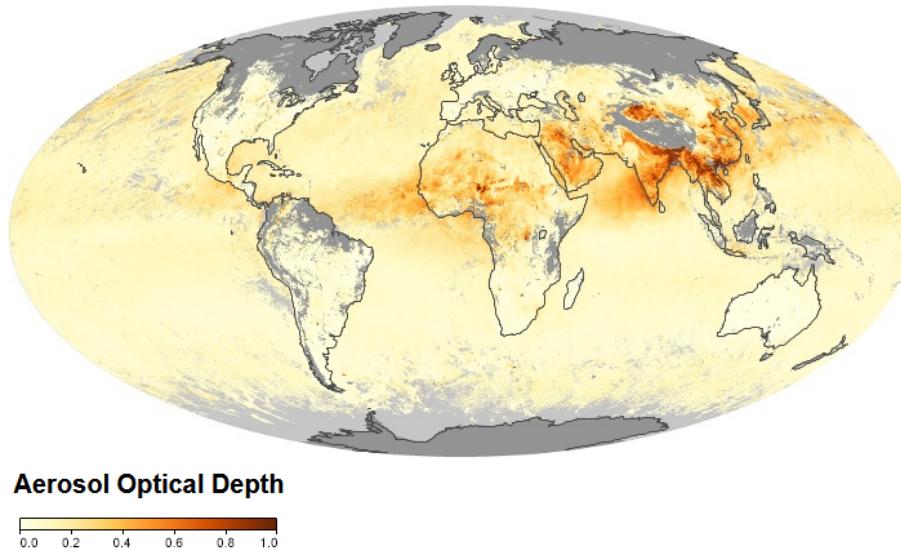


Figure 1: Aerosol optical depth from Terra MODIS, April 2025

Terra satellites. Both satellites orbit Earth from pole to pole each day, ensuring coverage of the entire world. Figure 1 displays a map of MODIS AOD data taken from the Terra satellite in April 2025, with darker pixels being indicative of greater levels of particulate matter (NASA (2025)). Although the spatial resolution of this image is quite low, it illustrates where airborne particulate matter is the most concentrated in the world. However, it is important to note that this image alone does not paint a full picture, as aerosol concentrations tend to fluctuate throughout the year due to the timing of certain agricultural practices or meteorological phenomena such as sandstorms or rainstorms.

Another satellite data source I encountered when researching this topic is the Earth Probe Total Ozone Mapping Spectrometer (TOMS), a satellite instrument featuring an aerosol index that tracks well with ground-based measures (Jayachandran (2009)). Consistent and accurate pollution measurements from instruments such as TOMS are especially useful when studying geographical regions that lack ground-based pollution monitoring systems. Some additional sources of non-AOD data include the Atmospheric Infrared Sounder (AIRS) on NASA’s Aqua satellite and the Ozone Monitoring Instrument (OMI) on the Aura satellite. I will discuss these satellite data sources more in section IV, where I attempt to use them in an empirical example.

3 Literature Review

In this section, I discuss a few academic papers that have taken advantage of access to remotely-sensed measures of pollution. Although my primary interest

in writing this paper is to explore how such measures can be treated as outcome variables to study the impacts of environmental policies, it is important to note that their use as explanatory variables highlights how expanded access to pollution data has facilitated social scientists' ability to study environmental phenomena on a global scale.

3.1 Satellite Pollution Data as an Explanatory Variable

As mentioned earlier, a key advantage to using satellite measures to study airborne pollution is the ease with which data can be collected at many points in time. Jayachandran (2009) exploits this ease of access to study the effects of wildfires in Indonesia in 1997, collecting daily satellite measurements to estimate how increases in airborne particulate matter attributed to the fires impacted infant mortality. Jayachandran collected her data on airborne smoke using the TOMS aerosol index. Ultimately, Jayachandran estimates a 20% increase in under-three mortality attributed to spikes in airborne pollution from the fires, with the estimated effect size being much larger in poorer areas. It should be noted that ground monitor data on air pollution in Indonesia were not available during the period of interest, which motivated the use of satellite data to study this particular topic. Without the global coverage of satellite data allowing for proxies of pollution even in undeveloped areas, a study like Jayachandran's may not have been feasible.

3.2 Satellite Pollution Data as an Outcome Variable

In their paper titled "Voluntary compliance, pollution levels, and infant mortality in Mexico," Foster et al. (2009) evaluate Mexico's Clean Industry Program, which is the main voluntary pollution reduction program in the country. In summary, the program allows plants to pay for an optional audit of their operations, after which they can receive a Clean Industry Certificate if they meet certain pollution emissions standards. The authors note that while previous literature has attempted to evaluate the impact of similar programs, a lack of a concrete measure of air quality has limited researchers' ability to evaluate the Clean Industry Program, motivating the use of satellite-based measures. Thus, using AOD data collected from Terra MODIS, the authors attempt to estimate the program's effects on pollution and infant mortality by using variation in local competition between auditors as an instrument for certification. To summarize their results, the authors estimated that the change in the proportion of firms certified by the program from 2000 to 2006 is associated with statistically significant decreases in AOD and respiratory infant mortality that are robust to different model specifications.

3.3 Limitations of Using Aerosol Optical Depth to Approximate Pollution

An important consideration for using AOD data, noted by Foster et al. (2009), is that discrepancies between AOD measurements and actual particulate matter may arise due to variation in meteorological factors. To attempt to account for this, they controlled for factors such as the dew point and temperature, both of which were measured from weather stations throughout Mexico, when running their regressions. This implies that AOD data alone may not always be sufficient when studying the effects of policies on levels of particulate matter or any other specific type of pollutant in a region of interest. As a result, challenges may arise when neither ground monitoring data nor weather stations are present in a given region, since variation in meteorological factors may create unwanted noise in AOD measurements and weaken their ability to proxy for pollution.

As mentioned above, AOD data on its own might only be valid as a proxy for specific types of pollution since meteorological factors outside of pollution contribute to the final estimates. As an example of how researchers have attempted to account for these factors, in their paper on the effects of Chinese air pollution policies from 2005 to 2017, Ma et al. (2019) take advantage of China’s recently established $PM_{2.5}$ (airborne particulate matter with a diameter less than $2.5\mu\text{m}$) ground monitoring network and use it to train a model they eventually use to estimate actual levels of $PM_{2.5}$ using historical AOD data collected from Aqua MODIS as well as data on meteorological factors and land use. The motivation behind this was that $PM_{2.5}$ data were not available in China before 2012, which is when the country first started considering $PM_{2.5}$ when implementing air pollution control policies. However, due to the accessibility of AOD data and satellite data in general, the researchers were able to generate fairly accurate predictions of actual particulate matter when also accounting for meteorological factors. Overall, studies such as this one illustrate that although AOD data alone may not be suitable for measuring specific types of pollutants, it certainly has merit in training models to predict pollution levels in periods of time when ground monitoring systems were unavailable.

Another potential issue with using AOD data is the possibility of measurement error, which may introduce bias in models such as the one from Jayachandran (2009). If AOD data suffers from classical measurement error, estimates of the effect of pollution on health outcomes using AOD as an explanatory variable will be attenuated. A related issue regarding measurement for satellite measures of pollution in general is the possibility of missing data at a given time due to cloud cover or other factors, which I will discuss in the next section.

4 Empirical Example: Jakarta, Indonesia’s ”Three-in-One” Policy

4.1 Overview of the Policy

Jakarta, Indonesia’s ”three-in-one” high occupancy vehicle (HOV) policy was a 1992 initiative to reduce traffic on the city’s main arteries during morning and evening rush hours. It required that all cars on the main streets of the city’s central business district carry at least three people during these hours. For context, Jakarta has the worst traffic in the world in terms of ”stops and starts”, with the average driver in Jakarta experiencing over 30,000 stops and starts per year. This figure is about twice as high as that for drivers in New York City, which has the worst traffic in the United States according to this metric (Hanna, Kreindler, and Olken (2017)).

In March 2016, the Jakarta government abruptly announced the abolition of its three-in-one policy, effective on April 5th that year. In their paper on the effects of this policy change on traffic delays, ”Citywide effects of high-occupancy vehicle restrictions: Evidence from ”three-in-one” in Jakarta,” Hanna, Kreindler, and Olken (2017) find that the abolition of the three-in-one policy resulted in increased delays throughout the city during most hours of the day, even on roads that were never subject to HOV restrictions.

4.2 Research Objective

My objective in this empirical section is to evaluate the degree to which satellite pollution data compares with ground-based pollution measures when evaluating the environmental impacts of the disbandment of the three-in-one policy. Since traffic within Jakarta was estimated to increase with the abolition of the three-in-one policy, and vehicles are known to be emitters of pollutants such as particulate matter, carbon monoxide (CO) and nitrogen dioxide (NO₂) (California Air Resources Board (2025)), this situation provides an opportunity to test whether satellite and ground-based measures of pollution changed by similar amounts, if at all, as a result of the policy change.

4.3 Data

4.3.1 Data Collection

To conduct my analysis, I collected data from three sources. First, I obtained data on daily measurements of various pollutants in Jakarta from ground-monitoring sources, which had been posted on Kaggle, a website where users can share datasets to be used for exploratory analysis. The dataset contains measures of the concentrations of various pollutants in the air in Jakarta between 2010 and 2025, such as particulate matter, CO, NO₂, sulfur dioxide, and ozone (Pohan (2025)).

Next, I collected daily measurements of AOD, NO₂, and CO using NASA's Giovanni data explorer, which allowed me to easily download these data for Jakarta from March 31st, 2016, to May 9th, 2016, from three separate satellite data sources:

1. The Atmospheric Infrared Sounder (AIRS) on NASA's Aqua satellite was used to collect measurements of the mole fraction of CO in the air during the daytime as the satellite crossed the equator from South to North in its orbit. The full name of this satellite data product is "Aqua/AIRS L3 Daily Standard Physical Retrieval (AIRS+AMSU) 1 degree x 1 degree V7.0 (AIRX3STD)"
2. MODIS on NASA's Terra satellite was used to collect AOD data. The name of the product is "MODIS Level 3 Atmospheric Gridded Product."
3. The Ozone Monitoring Instrument (OMI) onboard the Aura satellite was used to obtain measurements of the total column of NO₂ in the region. The product's name is "OMI/Aura NO₂ Cloud-Screened Total and Tropospheric Column L3 Global Gridded 0.25 degree x 0.25 degree V3 (OMNO2d)."

Lastly, I downloaded replication data from Hanna, Kreindler, and Olken (2017), which contains data on Google Maps API queries performed every 10 minutes from March 31st, 2016, to May 9th, 2016. This dataset contains variables related to current travel conditions at the time of each query, such as Google Maps' predicted travel duration routes on former three-in-one roads, as well as roads that were never subjected to the policy. In an attempt to perform a simple robustness check of the original study's findings, I averaged expected travel times for each day and included these averages as a variable in the final dataset I used for my analysis. As a result, my final dataset contains the following variables:

1. *duration*: A measure of travel duration, accounting for traffic, collected by Hanna, Kreindler, and Olken (2017) using the Google Maps API. I averaged its values by day to match the temporal resolution of my other variables.
2. *co_sat*: The satellite measure of carbon monoxide collected from AIRS
3. *co*: Carbon monoxide measured at ground-monitoring stations in Jakarta
4. *no2_sat*: The satellite measure of nitrogen dioxide collected from OMI
5. *no2*: Nitrogen dioxide measured at ground-monitoring stations in Jakarta
6. *AOD*: The aerosol optical depth measures,

4.3.2 Data Limitations

Figure 2 compares the actual border of Jakarta to the bounding box I selected that depicts the region in which satellite data was downloaded for the period of interest. Although my bounding box does not match the exact border of Jakarta, I chose a square region that roughly contains the entire city in the hope that it would capture any pollution coming directly from vehicles during the day. It is important to note that the bounding box I selected has dimensions of $0.5^\circ \times 0.5^\circ$, making it smaller than what is permissible for the AIRS and MODIS products, which provide measurements in $1^\circ \times 1^\circ$ grid cells. When this is the case, the Giovanni tool will collect data from the nearest available $1^\circ \times 1^\circ$ grid cell. This means that my measurements of AOD and the mole fraction of CO in the air were estimated from a single pixel on a given day. However, the OMI product has a spatial resolution of $0.25^\circ \times 0.25^\circ$, implying that estimates of the NO₂ column for a given day were averaged across 4 pixels of measurements.



Figure 2: Left: Jakarta, Indonesia on Google Maps. Right: Bounding box at 106.5E, 6.5S, 107E, 6S.

Upon inspecting the AOD data for Jakarta during the time period of interest, I came to the realization that it would be practically unusable, as it contained missing values for 29 out of 40 days, which is illustrated in Figure 3. This was not as much of an issue for the CO and NO₂ data, which were missing 10 and 12 days' worth of measurements, respectively. However, considering that the original sample size of 40 days was already very small, this reduction in usable data severely limited its ability to be used to produce any insightful analysis. The only silver lining in the case of the CO and NO₂ data was that they never contained missing values on consecutive days, which means that they might be well suited to using interpolation to fill in the missing values if one desired to preserve the original sample size.

One plausible explanation for why missing data were so prevalent in my satellite measures could be the presence of cloud cover preventing any mea-

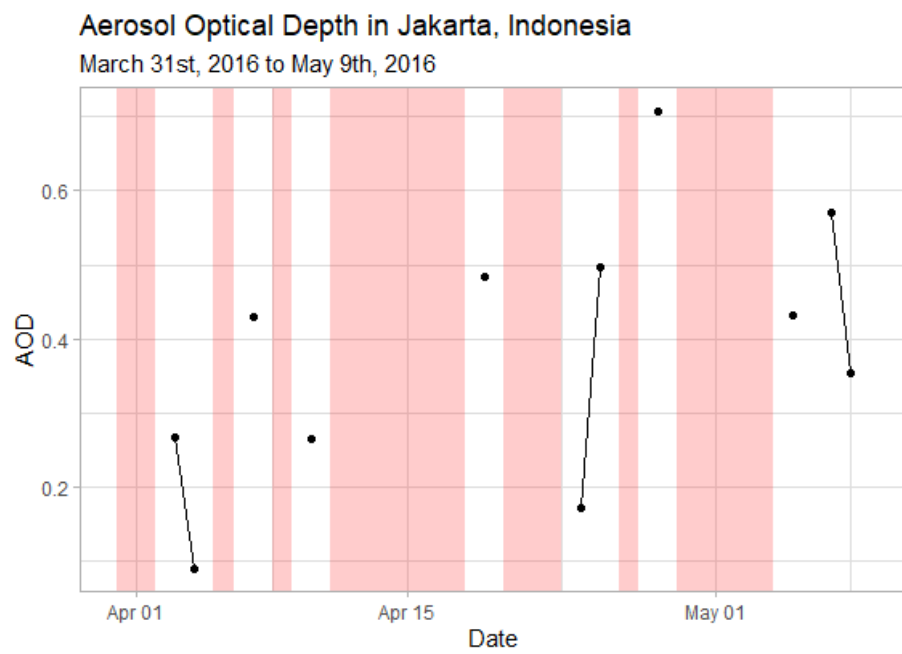


Figure 3: Missing AOD Data: Stretches of missing data are given by the red bands.

measurements from being recorded at all. For instance, the NO₂ data only contain measurements for days in which cloud cover made up less than 30% of the region of interest. Since the spatial resolutions of the satellite measures I used were quite low, coupled with the fact that my bounding box was very small, it is likely that the measures were extremely sensitive to the presence of any amount of cloud cover in Jakarta during this time period. While this might not explain all instances of missing data in the measures of AOD and CO, it offers insight into the complications that may arise from attempting to use satellite data to study pollution outcomes in such small geographic areas. Thus, satellite data may be more useful for studying more widespread environmental phenomena, such as the wildfires studied by Jayachandran (2009).

4.4 Empirical Strategy

In order to compare changes in satellite pollution measures after the policy change with those experienced by ground-based measurements, I employed a similar approach to Hanna, Kreindler, and Olken (2017), running first-difference regressions of my outcomes of interest on an indicator for whether a given day took place before or after the policy change on April 5th. Thus, my regression specification is the following:

$$\log(y_d) = \beta_0 + \beta_1 \times post_d + \epsilon_d,$$

where y_d represents the pollution measure of interest measured on day d ; β_1 is the coefficient of interest, measuring the difference in the average measurement of a given pollutant in Jakarta after the policy was lifted compared to before; and $post_d$ is an indicator for whether a given day takes place before or after the policy change. The main assumption in using the days before the policy as a counterfactual is that pollution in Jakarta would not have experienced notable changes during this period of time had the policy been kept in place. To account for discrepancies in the units of each pollution measure, I took the natural log of each outcome variable when running my regressions, which would ensure that the estimated coefficients could be interpreted as percentage changes.

4.5 Results

Table 1 illustrates the results of my first-difference regressions of traffic duration and pollution on an indicator for whether a given day took place after the policy change on April 5th, 2016. The most glaring problem with my results is that none of my estimates of the effect of the policy change on pollution or traffic duration are statistically significant. This is most likely because my sample size of 40 days is incredibly small, which will inflate the standard errors of the OLS estimates. This issue could be partially addressed by using a higher temporal resolution, such as hourly measures as opposed to daily, during the data collection process. However, I was limited in that my ground-based pollution monitoring data only contained daily measurements, and my goal was to

Table 1: First-Difference Regressions

	<i>Dependent variable:</i>				
	log(duration)	log(co_sat)	log(co)	log(no2_sat)	log(no2)
	(1)	(2)	(3)	(4)	(5)
post	0.051 (0.074)	0.009 (0.061)	0.332 (0.259)	-0.037 (0.209)	0.214 (0.322)
Constant	6.246*** (0.069)	4.628*** (0.057)	2.966*** (0.242)	36.030*** (0.194)	1.970*** (0.305)
Observations	40	30	40	28	39
R ²	0.012	0.001	0.042	0.001	0.012
Adjusted R ²	-0.014	-0.035	0.016	-0.037	-0.015

Note:

*p<0.1; **p<0.05; ***p<0.01

compare the effects of the policy change on these measures with the effects of satellite measures. Another factor that may have reduced the statistical power of my estimates is the fact that the spatial resolution of my satellite pollution measures was quite low, which would have likely made them more sensitive to slight fluctuations in pollution or other meteorological factors. Such noisy pollution measurements might also contribute to inflating the standard errors of the OLS estimates when considering these measures as outcome variables.

While my estimates should be taken with a grain of salt due to their lack of statistical significance, it is worth noting that besides the satellite measure of NO₂, all of the other outcomes of interest were estimated to have increased after the policy change, given by the coefficients on *post* being positive. However, assuming that traffic was the main driver of changes in NO₂ or CO pollution in Jakarta during this period of time, it is surprising that the estimated magnitude of the effect of the policy change on trip duration was quite low, around a difference of 5%. It would be interesting to see if these trends persist when using data that has been collected at more frequent time intervals.

5 Conclusion

In the first half of this paper, I explored how remote sensing techniques have facilitated researchers' abilities to study pollution on a global scale. In general, the public availability of satellite-based pollution measures has made it incredibly easy to access pollution data throughout the world for the periods of time during which they have been available. However, in the second half of my paper, where I discussed an empirical example related to an HOV policy in Jakarta, Indonesia, I realized first-hand that satellite pollution data do not come

without drawbacks. Unfortunately, the measures I obtained did not appear to be well-suited to studying pollution in such a small geographical region over a short period of time. However, it is worth noting that even the ground-based measures of pollution I obtained did not offer much insight into how pollution may or may not have changed within Jakarta as a result of the abolition of the HOV policy. It is entirely possible that pollution did not change significantly as a result of this policy change; however, my lack of conclusive results may also be a consequence of my inexperience in developing and executing a research design. If I were to expand upon what I have done in the future, it may be beneficial to consider an entirely different policy impacting a broader geographic region over a longer time interval, which may overcome the pitfalls I discussed in my empirical section.

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